

NHA TRANG, Viet Nam

WMO: 488770

Lat: 12.228N

Lon: 109.192E

Elev: 33

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.4	70.0	59.8	76.9	71.6	61.7	82.4	72.6	25.3	76.0	23.3	76.2	3.0	290	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.9	92.3	80.0	91.3	79.9	90.4	79.8	82.0	89.0	81.5	88.5	80.9	87.9	7.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.5	86.9	79.5	153.1	86.4	78.9	150.3	86.1	46.0	89.4	45.3	88.8	44.7	88.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.2	18.5	16.2	DB	64.0	96.5	4.4	2.8	60.8	98.5	58.2	100.1	55.8	101.7	52.6	103.7
			WB	60.4	83.6	3.1	0.9	58.2	84.2	56.4	84.7	54.6	85.2	52.4	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.2	76.1	77.1	79.6	82.6	84.4	84.9	84.5	84.6	83.3	81.2	79.3	77.1	(6)
	DBStd	3.77	2.14	2.34	2.30	1.98	1.95	1.91	1.80	1.99	1.83	1.95	2.12	2.43	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11405	809	759	917	978	1065	1048	1071	1074	999	966	878	841	(10)
	CDD65	5930	344	339	452	528	600	598	606	609	549	501	428	376	(11)
	CDH74	61558	1884	2269	4025	5904	7347	7508	7440	7520	6329	5106	3717	2509	(12)
	CDH80	20640	115	273	864	1993	3079	3309	3124	3228	2361	1409	654	231	(13)
(14) Wind	WSAvg	5.1	7.5	6.9	5.7	4.4	3.5	3.2	3.2	3.2	3.2	4.6	6.6	8.9	(14)
(15) Precipitation	PrecAvg	65.9	1.9	0.4	1.4	2.7	4.3	2.8	1.9	3.4	8.6	14.3	15.9	8.3	(15)
	PrecMax	108.1	5.4	1.4	7.9	8.7	10.1	10.1	6.1	6.0	18.4	38.2	36.7	50.0	(16)
	PrecMin	37.9	0.4	0.0	0.0	0.7	1.3	1.1	0.0	1.1	3.3	6.7	3.7	0.2	(17)
	PrecStd	18.2	1.5	0.4	2.4	2.0	2.4	2.0	1.3	1.3	3.7	6.7	9.1	9.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	85.9	88.1	90.7	92.9	93.8	93.1	93.6	91.8	89.0	86.7	85.1	(19)
		MCWB	75.8	76.2	77.8	80.0	80.9	80.4	79.6	79.2	79.8	79.4	78.0	76.7	(20)
	2%	DB	82.4	84.3	86.6	89.4	91.4	92.1	91.4	91.7	90.5	87.7	85.6	83.6	(21)
		MCWB	74.8	75.9	77.4	79.4	80.6	80.1	79.5	79.5	79.8	78.9	77.8	76.1	(22)
	5%	DB	81.3	83.0	85.4	88.3	90.3	90.8	90.2	90.5	89.3	86.7	84.6	82.4	(23)
		MCWB	74.0	75.3	77.0	79.0	80.4	79.9	79.4	79.5	79.6	78.5	77.4	75.4	(24)
	10%	DB	80.2	81.7	84.3	87.2	89.2	89.6	89.1	89.4	88.2	85.7	83.5	81.2	(25)
		MCWB	73.3	74.6	76.5	78.8	80.2	79.7	79.2	79.4	79.2	78.2	76.9	74.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	78.1	79.4	81.7	82.8	82.8	82.2	82.6	82.1	80.9	80.0	78.7	(27)
		MCDB	81.9	83.4	85.3	88.0	89.3	89.7	89.5	89.4	88.8	86.7	84.5	82.8	(28)
	2%	WB	76.4	77.1	78.7	80.7	81.8	81.8	81.1	81.5	81.1	80.1	79.1	77.6	(29)
		MCDB	80.8	82.4	84.4	86.9	88.7	89.1	88.4	88.5	88.0	85.7	83.8	81.6	(30)
	5%	WB	75.4	76.3	78.1	80.0	81.2	81.1	80.5	80.8	80.5	79.4	78.3	76.8	(31)
		MCDB	79.7	81.4	83.6	86.1	88.2	88.4	87.7	88.1	87.4	85.1	83.0	80.7	(32)
	10%	WB	74.5	75.5	77.4	79.5	80.6	80.4	79.9	80.1	79.8	78.9	77.7	76.0	(33)
		MCDB	78.9	80.5	82.9	85.5	87.7	87.8	87.1	87.4	86.6	84.6	82.2	79.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.8	9.0	9.2	10.0	9.9	9.8	10.2	10.1	8.6	7.3	6.8	(35)
		MCDBR	9.1	9.8	10.1	10.1	10.9	11.0	10.8	11.2	11.4	9.9	8.9	8.5	(36)
	5% WB	MCWBR	5.1	5.2	4.9	4.5	5.1	4.9	4.8	4.8	5.3	4.8	5.0	4.7	(37)
		MCWBR	8.0	9.0	9.2	9.4	10.3	10.2	10.2	10.4	10.6	9.2	7.9	6.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.456	0.446	0.488	0.470	0.450	0.436	0.430	0.432	0.445	0.487	0.478	0.491	(40)
		taud	2.283	2.309	2.204	2.285	2.374	2.437	2.440	2.423	2.380	2.245	2.259	2.174	(41)
	Ebn at Noon		268	276	266	268	269	270	272	275	274	261	260	254	(42)
		Edh at Noon	42	42	48	44	39	36	36	38	39	44	43	46	(43)
(44) All-Sky Solar Radiation	RadAvg		1351	1685	1794	1957	1857	1752	1715	1709	1635	1402	1249	1128	(44)
	RadStd		142	147	168	101	128	99	110	130	134	144	173	189	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.68	+1.19	N/A	N/A	+0.61	+0.58	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	+0.45	+0.85	N/A	N/A	+0.49	+0.50	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page